Quantitative Methods in Geographic Research

Welcome and Introduction

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Outline

- Introductions
- Course Motivation and Overview
- Syllabus and Logistics

Introductions

- Name/Preferred Name (if different)
- Major & Year
- I (don’t) know what you did last summer!
  - A place you visited?
  - A dish you ate?
  - A hobby you picked up?
- You have $1 million and one-way ticket. Where would you want to settle in the next 5 years?

About Me

- ‘Junior’ Assistant Professor
- Health and Medical Geographer
  - Lots of GIS + Statistics
  - Designing studies for geographic and health research
  - Applied Policy work –> research translation
- First name pronounced as *‘Maroon’* with a *‘V’*
- Last name pronounced as *‘Go-yell’*
- You can call me Varun, Dr. Goel, Dr. G, Professor V

My research

- Global Environmental Health (Asia, Africa)
  - Climate Change and Health
  - Human-Environment (and animal) Interactions
- Infectious Disease Ecology
  - Malaria, Diarrhea, HIV, others
- Evaluation of Health interventions
- Applied Spatial Statistics

What is this course about?

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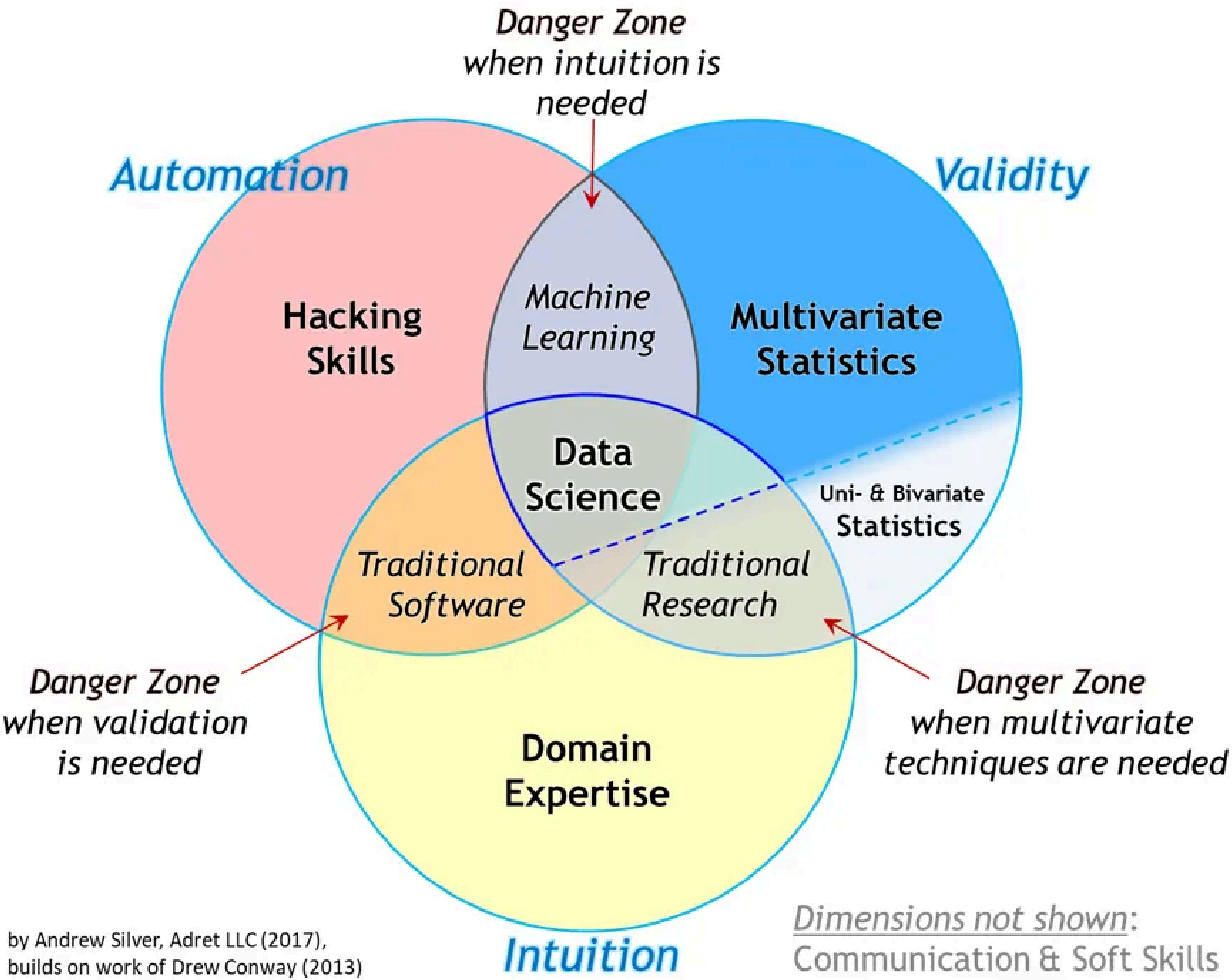

Danger Zone when intuition is needed
Automation
Validity
Hacking Skills
Machine Learning
Multivariate Statistics
Data Science
Uni- & Bivariate Statistics
Traditional Software
Traditional Research
Danger Zone when validation is needed
Danger Zone when multivariate techniques are needed
Domain Expertise
Intuition
by Andrew Silver, Adret LLC (2017), builds on work of Drew Conway (2013)
Dimensions not shown: Communication & Soft Skills

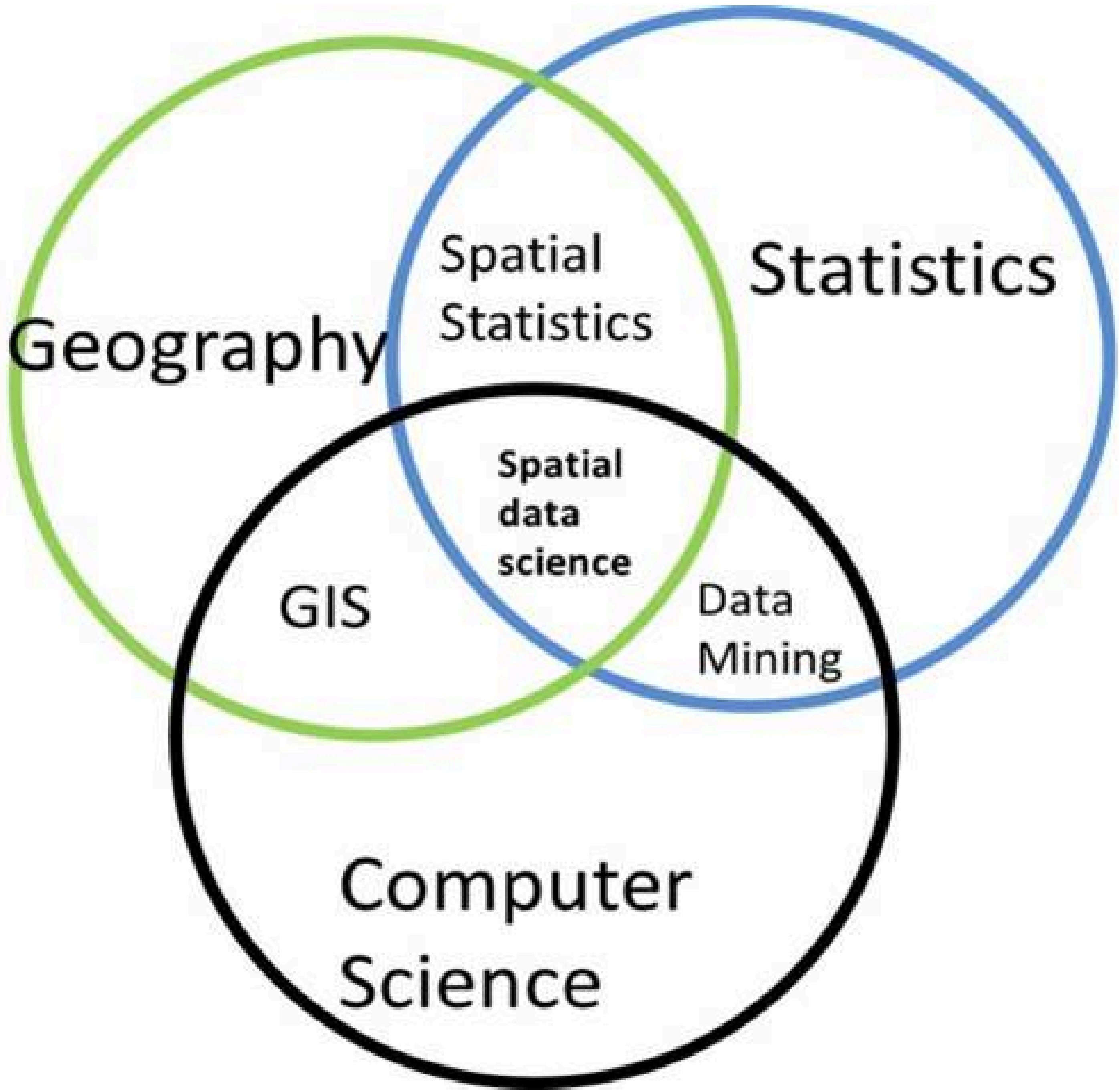
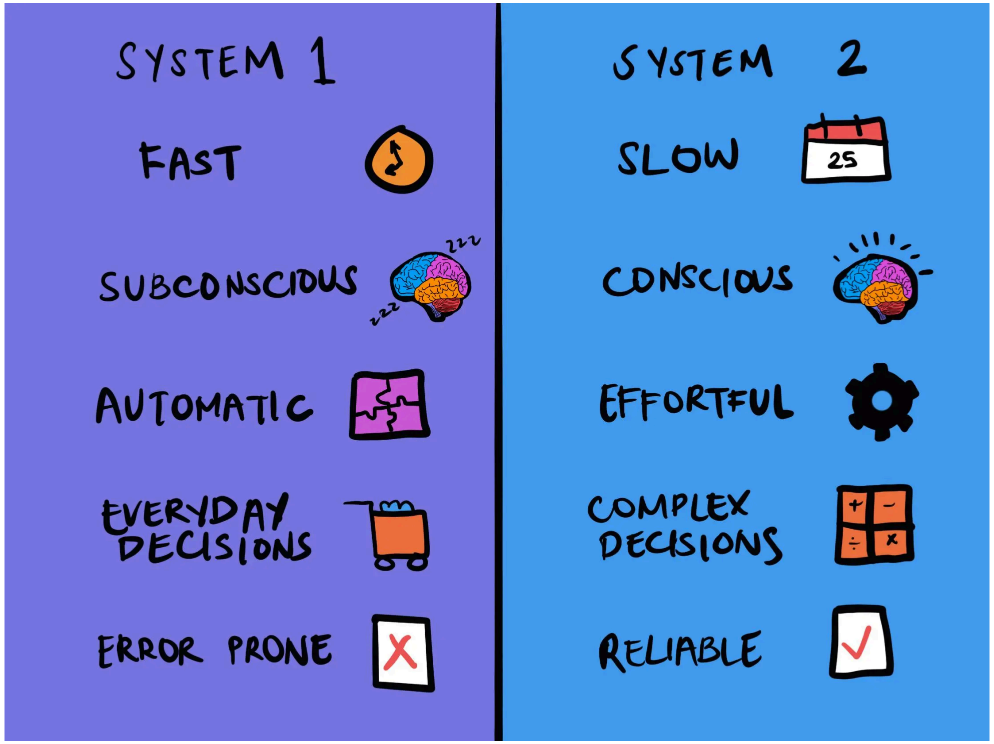

Geography
Spatial
Statistics
Statistics
Spatial
data
science
GIS
Data
Mining
Computer
Science

Figure 1. Seven components of quantitative geography.
It's about geography more than it is about maths or statistics
(Source: adapted from Harris, 2016)

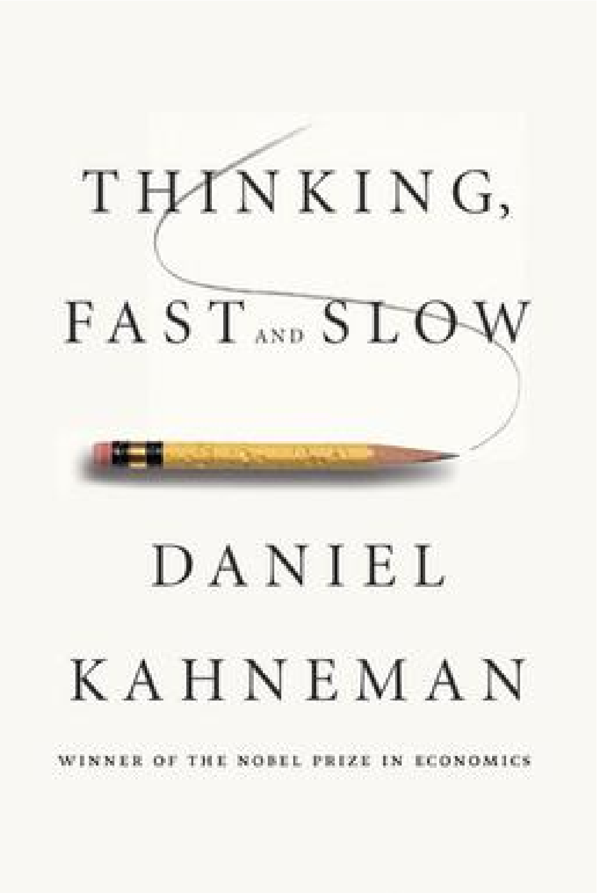
Quantitative Methods (thinking) in Geography

- Statistics
- Research methods/design
- GIS
- Spatial Analysis
- Programming
- Science Communication

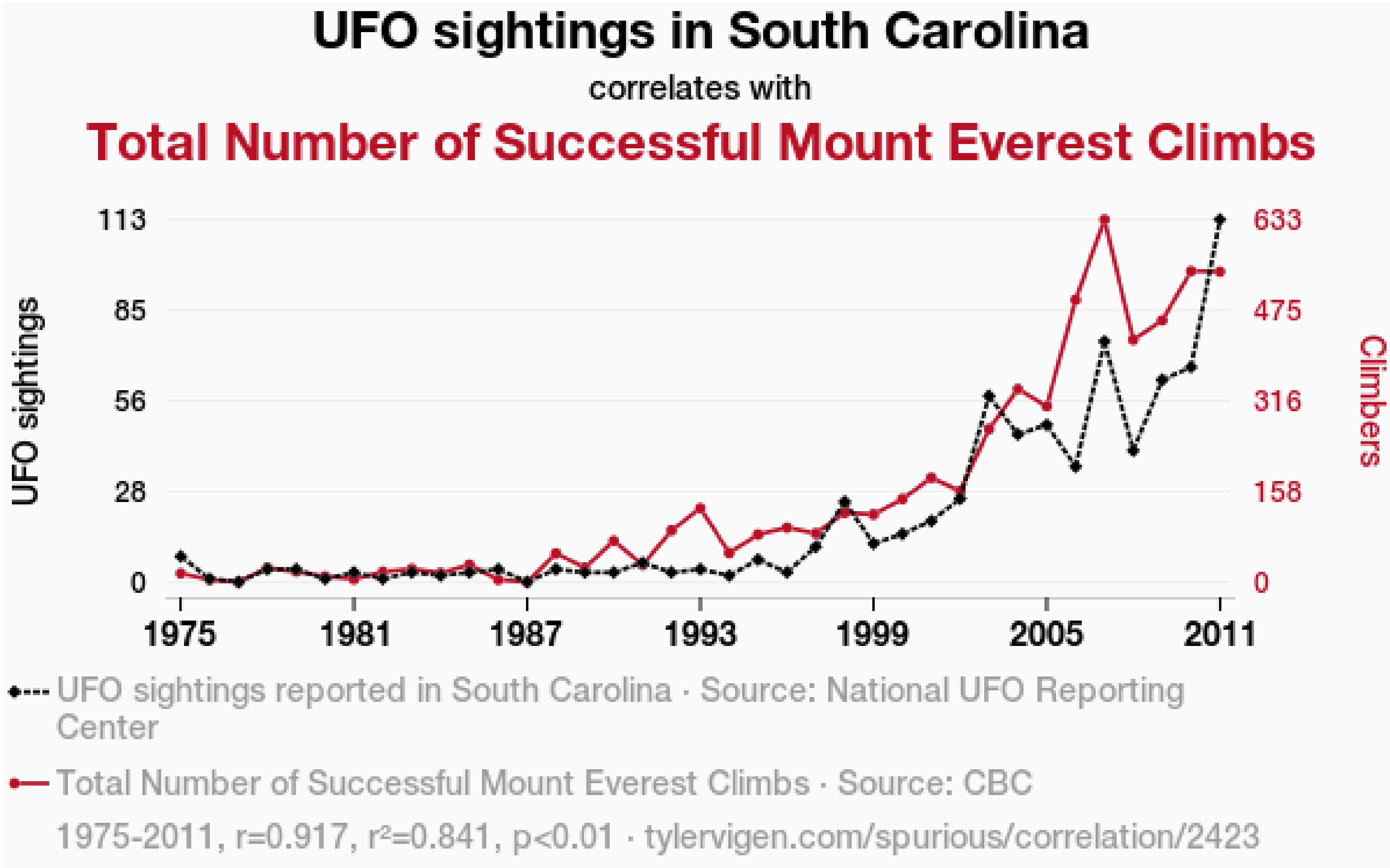
Statistics



Source: The Decision Lab



Research Design



Source: Tyler Vigen

Better and Reliable Science Communication

THEY STUDIED DISHONESTY. WAS THEIR WORK A LIE?

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

Example: Jo(h)n Snow and Cholera

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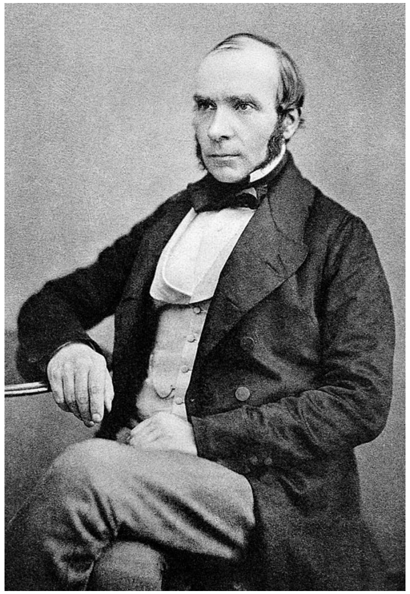
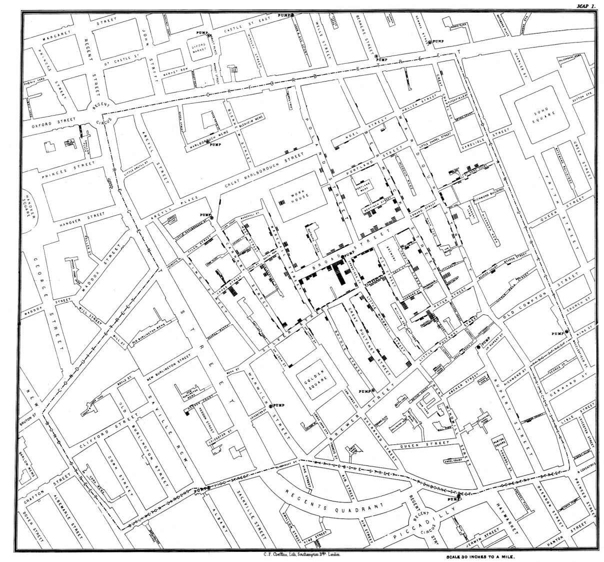
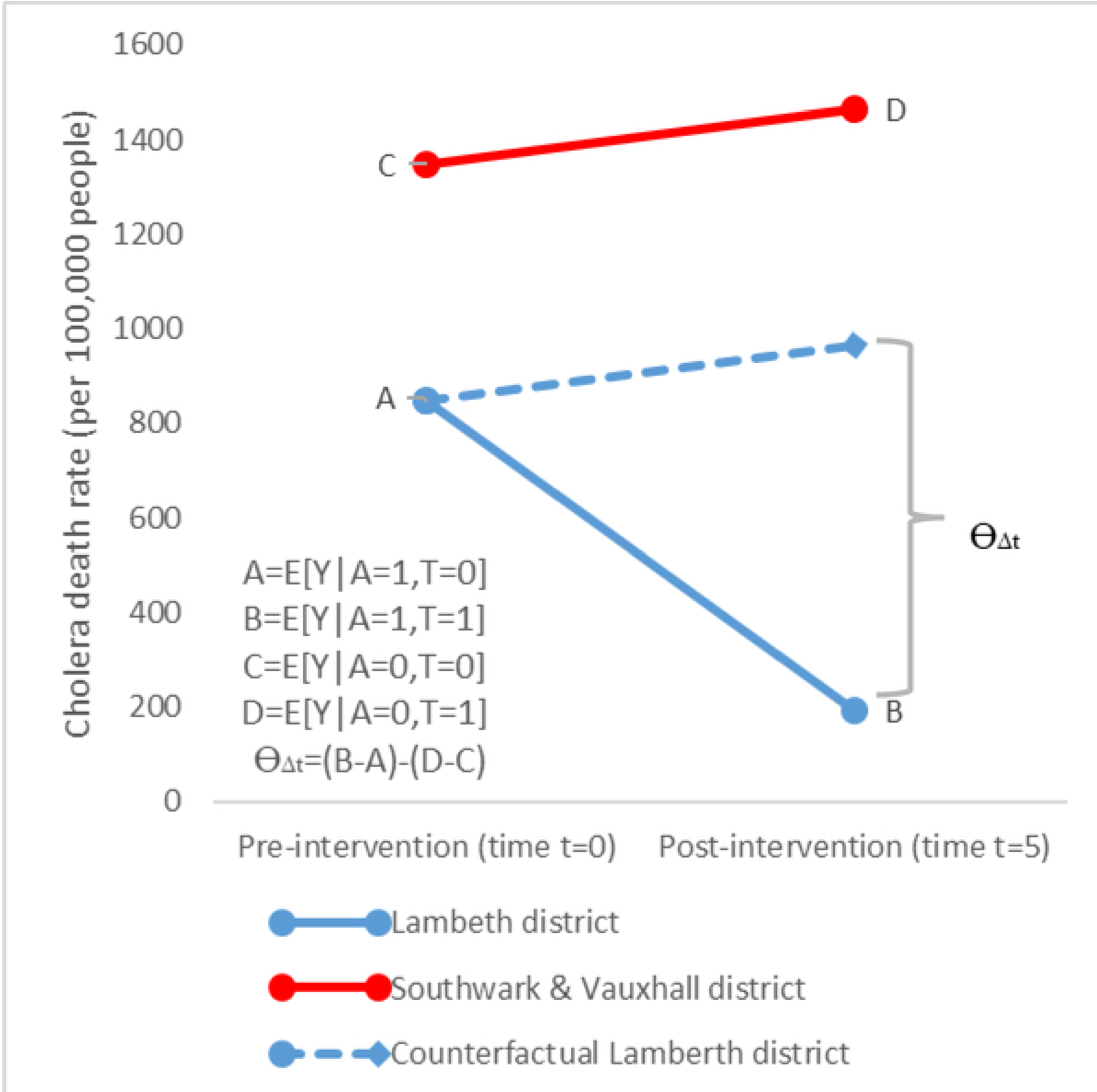
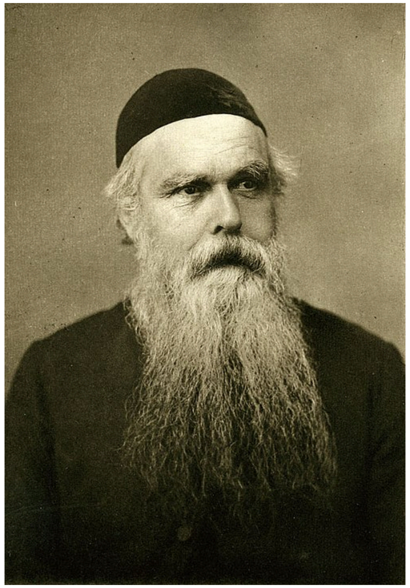




Example: Jo(h)n Snow and Cholera



Example: Jo(h)n Snow and Cholera



Example: Jo(h)n Snow and Cholera

TABLE III.

Shewing the results of the whole Inquiry during the First Seven Weeks of the Epidemic, arranged in Districts.

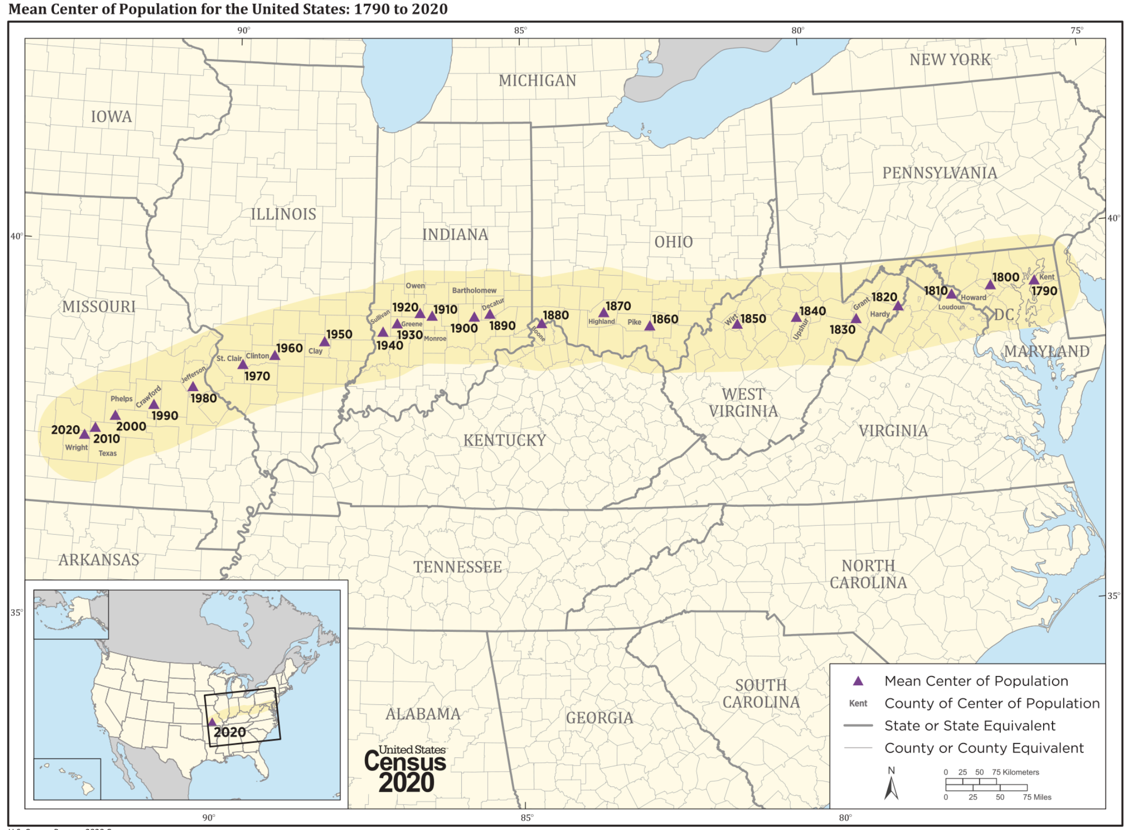
Registration Districts.	Number of inhabited houses in 1851.	Population in 1851.	Estimated constant population per house.	"Number of houses, and estimated number of persons, supplied in 1854 with water as under." By the Southwark and Vauxhall Co. No. of houses.	Estimated population	By the Lambeth Company. No. of houses.	Estimated population	Water supply of the houses in which fatal attacks of cholera took place during first seven weeks of epidemic of 1854. Southwark and Vauxhall Co.	Lambeth Co.	Thames, canals, or ditches.	Pump-wells.	Supply not ascertained.	Deaths in first seven weeks of epidemic.	Mortality per 10,000 supplied with water as under. Southwark and Vauxhall Co.	Lambeth Co.
St. Saviour, Southwark	4,600	35,731	7·8	2,631	19,617	1,089	14,201	126	13	10	0	1	150	64·4	9·1
St. Olave, Southwark	2,360	19,375	8·2	2,193	18,638	0	0	91	0	8	0	5	104	48·8	..
Bermondsey...............	7,007	48,128	6·9	8,402	57,884	268	1,785	266	0	25	0	0	291	45·9	..
St. George, Southwark	6,992	51,824	7·4	3,419	25,039	3,183	23,712	134	20	0	0	3	157	53·5	8·4
Newington	10,458	64,816	6·2	5,224	31,940	5,473	33,531	155	11	0	1	2	169	48·5	3·2
Lambeth	20,447	139,325	6·8	8,077	54,982	11,763	83,786	176	43	8	7	9	243	32·8	4·9
Wandsworth	8,276	50,764	6·1	3,028	18,390	618	3,870	62	1	16	17	0	96	33·7	3·3
Camberwell	9,412	54,667	5·8	4,005	23,472	1,835	10,478	185	9	0	2	1	197	78·8	8·5
Rotherhithe...............	2,792	17,805	6·4	2,336	14,951	0	0	68	0	35	0	0	103	45·4	..
Sub-district of Sydenham....	801	4,501	5·6	0	0	unknow.	unknow.	0	1	0	2	1	4	..	..
Not identified	..	..	6·6	411	2,712	25	165	..	..	..	..	..	..	..	..
Totals...............	73,145	486,936	6·7	39,726	267,625	24,854	171,528	1263	98	102	29	22	1514	47·2	5·7

IN THE SOUTH DISTRICTS OF LONDON.

253

A simple yet powerful descriptive spatial statistic

Mean Center of Population for the United States: 1790 to 2020



Source: Wikipedia/US Census

Logistics

- **Syllabus**
- **Attendance and Participation**
- **Use of Generative AI**.
- **Hosting materials.** Blackboard v/s Course website
- **Asking for Help**. Be Proactive

R

- Programming Language written by Statisticians primarily for statistical computing
- Big community and support
- Suitable for Geographic Research
    - Very strong spatial statistics capabilities
    - Great data visualization + mapping
    - Supported by packages (similar to libraries in Python)

R Studio, Posit Cloud

- R Studio: Integrated Development Environment for R
  - Makes Dull R more Fun
- Posit Cloud: Rstudio in the cloud
  - Reduces steep learning curve
  - Reduces technical issues
- Sign up at https://posit.cloud/ with your SCHOOL email

Questions/Concerns?